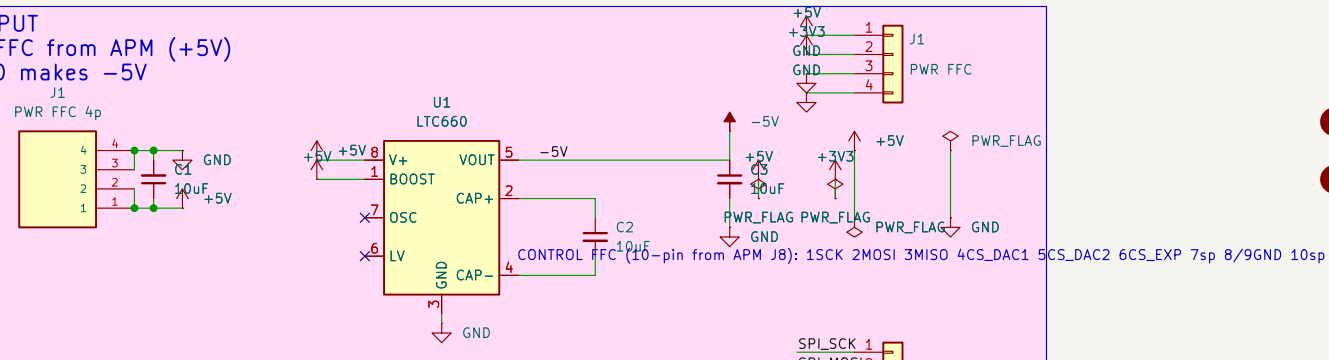
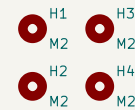


POWER FFC (4-pin from APM): +5V +3V3 GND GND (-5V made on-board)

- POWER INPUT
 - 4-pin FFC from APM (+5V)
 - ICL7660 makes -5V

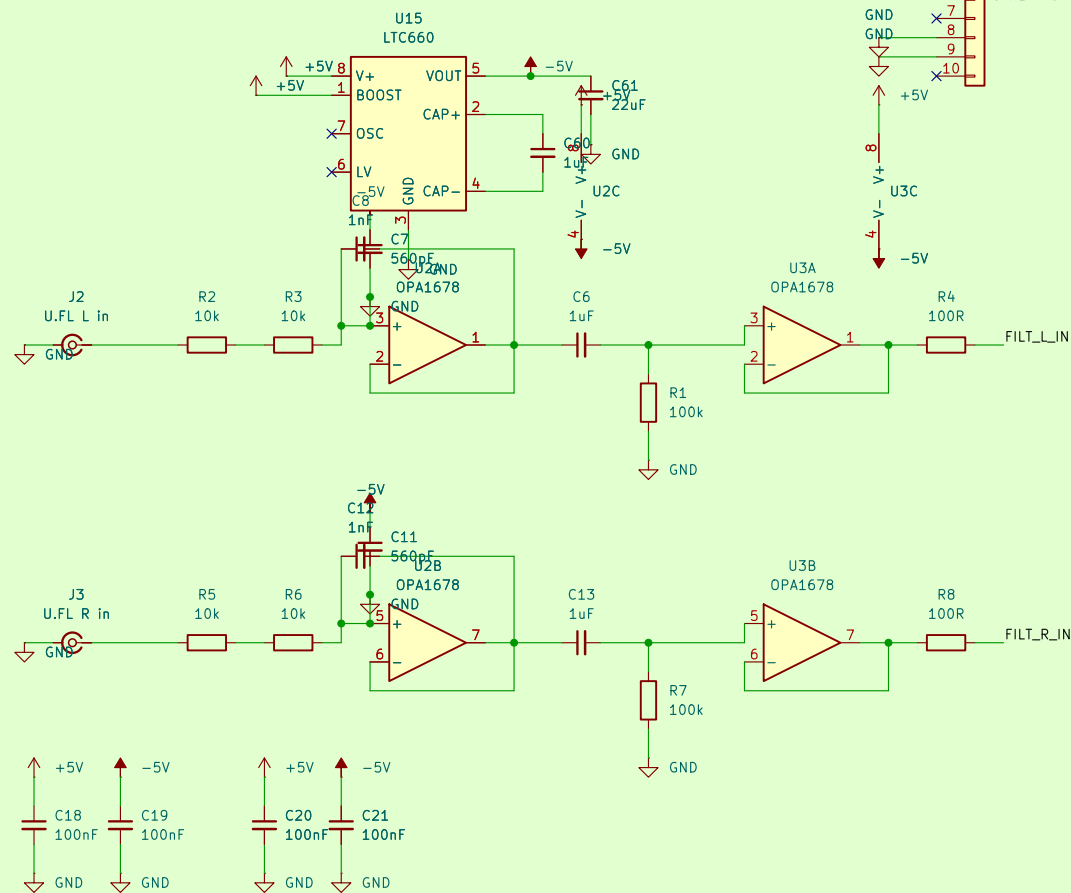


OUTPUTS - 1/4in L/R (ACJS-MHOM) + 3.5mm headphone/debug

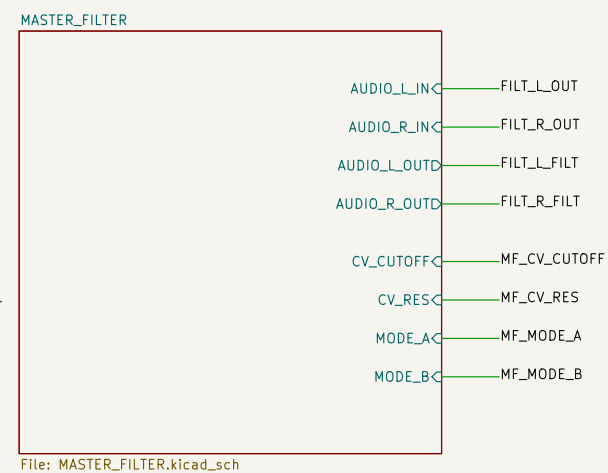
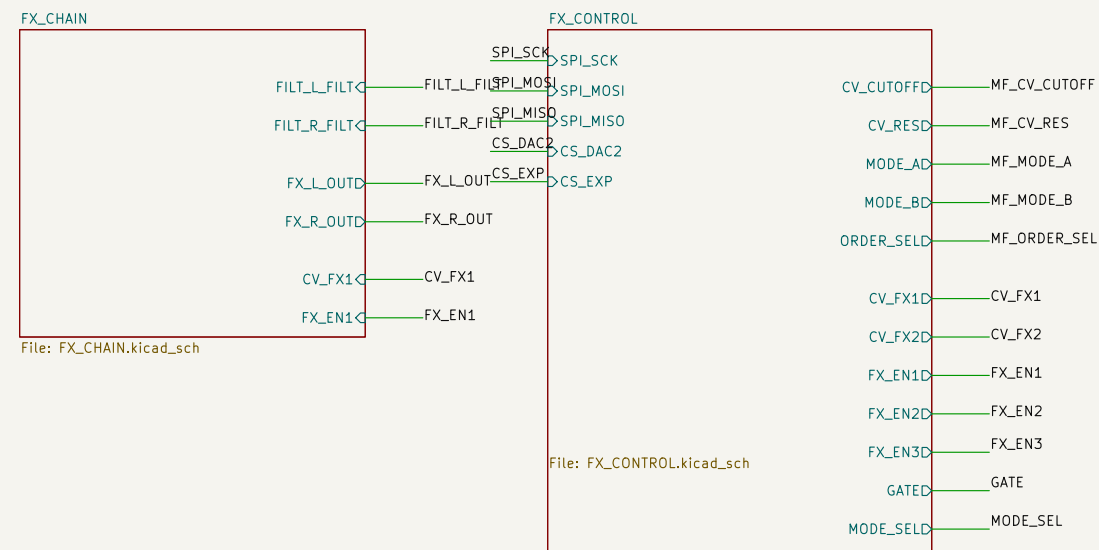
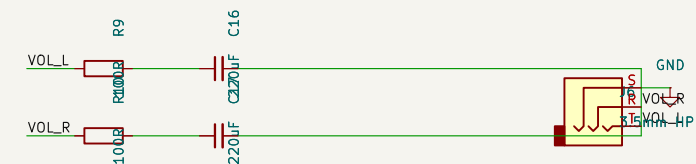
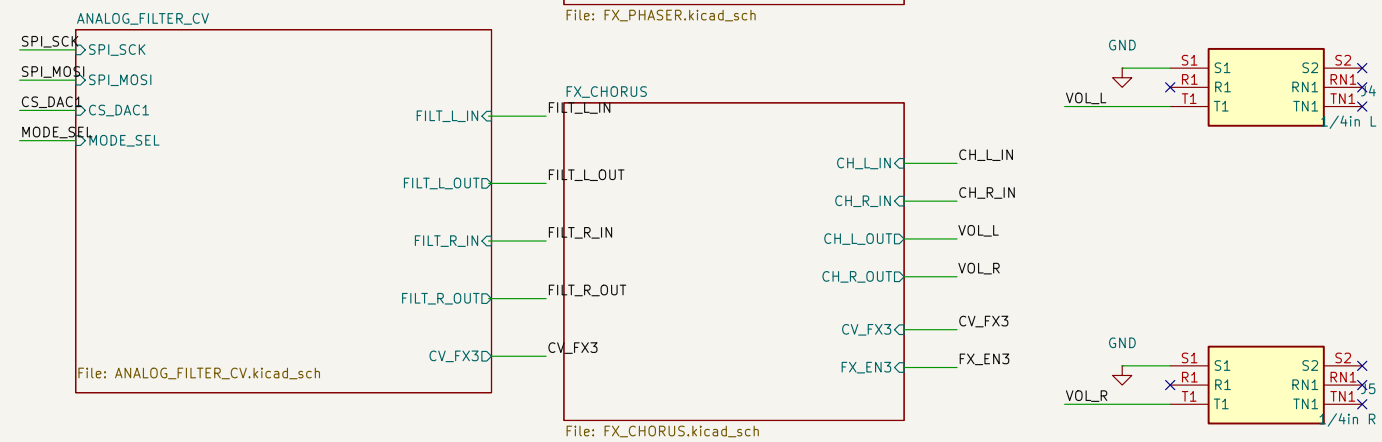


AUDIO INPUT

- U.FL for L and R DACs from APM



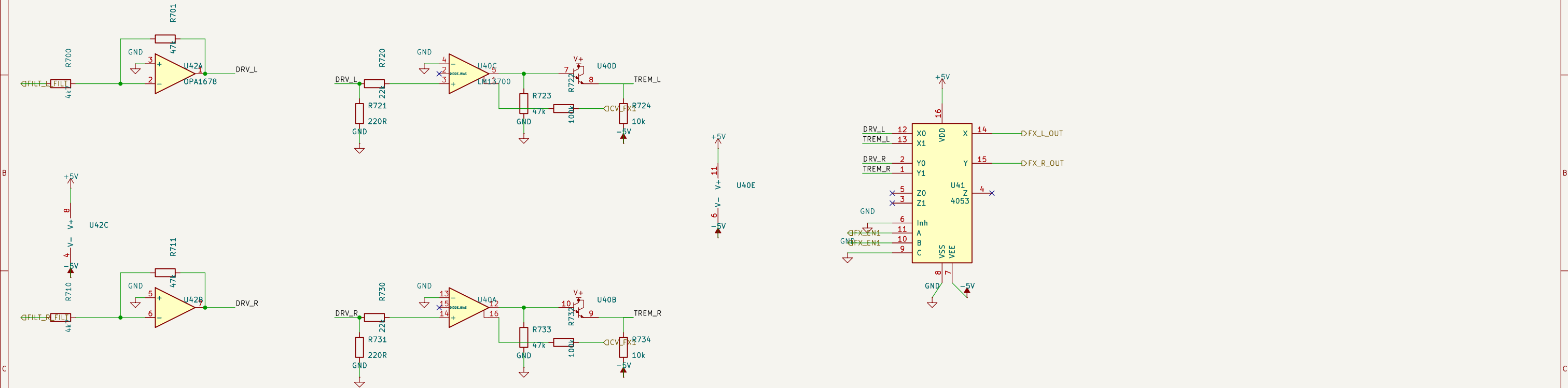
ANALOG PATH – each channel's buffer output (FILT_x_IN) goes into the analog sub-sheet; FILT_x_OUT returns to the jacks



MASTER VOLUME – RV4/RV5 = one dual-gang pot (shared L/R knob); attenuates final output to the jacks/HP



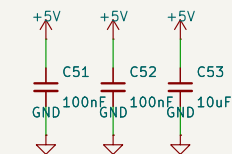
FX CHAIN - ¹DRIVE (soft clip) + ²TREMOLO (OTA VCA, LFO=CV_FX1); each bypassable (FX_EN). In: ³FILT_x_FILT (from ⁴SVF) -> ⁵Out: FX_x_OUT (to volume) ⁶ ⁷ ⁸



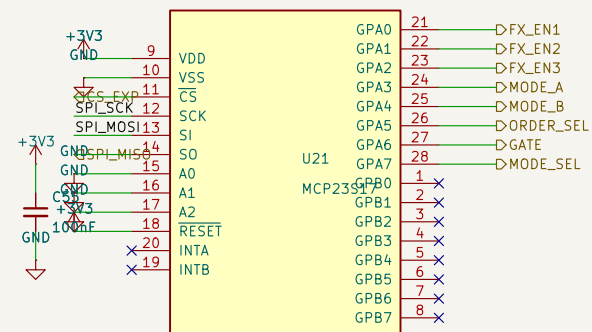
NOTE: analog FX – simulate/verify diode clip levels and OTA VCA vs LM13700 datasheet before fab.

NOTE: analog FX – simulate/verify diode clip levels and OTA VCA vs LM13700 datasheet before fab.

Sheet: /FX_CHAIN/ File: FX_CHAIN.kicad_sch		
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2 (-> future phaser/chorus/tremolo); MODE_A/B, ORDER_SEL, FX_EN1/2 digital.



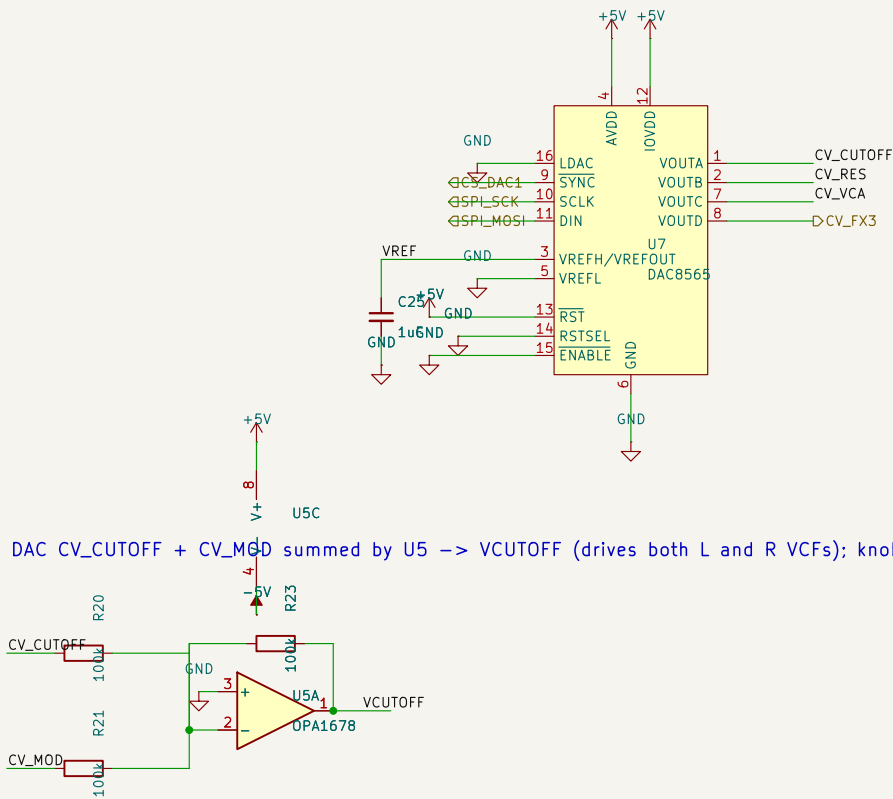
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File: FX_CONTROL.kicad_sch		
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Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 3/7

ANALOG FILTER / CV (stereo) – SPI in (10-pin FFC) -> DAC8565 makes CVs -> per-channel LM13700 VCF/VCA; CD4053 selects analog or FPGA-digital

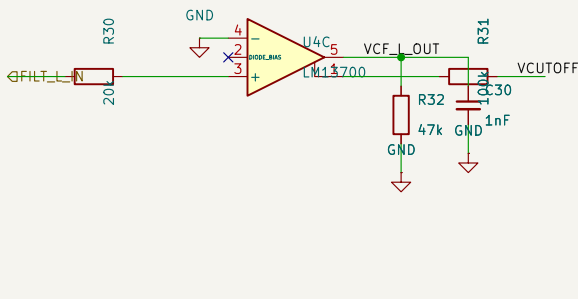
CONTROL/DAC – one 10-pin control FFC (root) feeds this DAC (CS_DAC1) + the FX DAC + MCP over ONE shared SPI bus; MODE_SEL from the expander

TOGGLE – CD4053: MODE_SEL selects analog (VCA_x_OUT) vs bypass (FILT_x_IN) for BOTH L and R

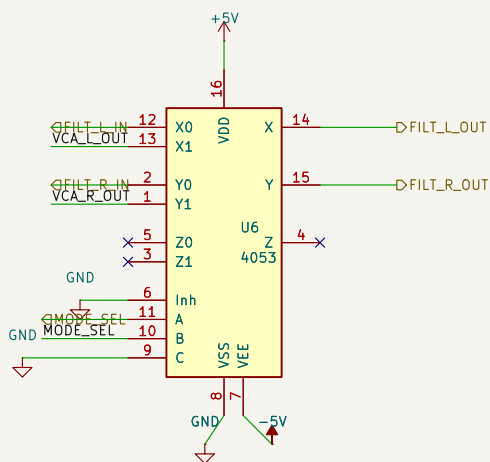
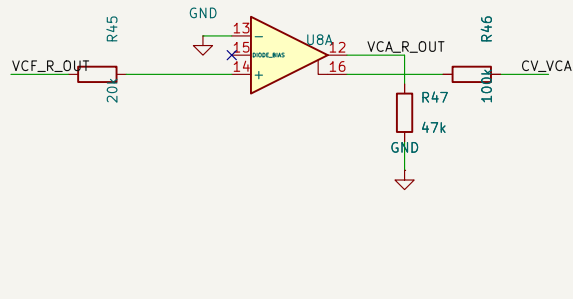
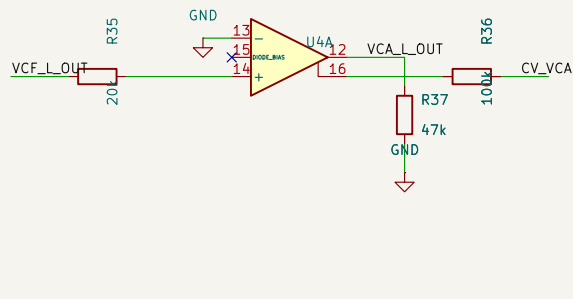
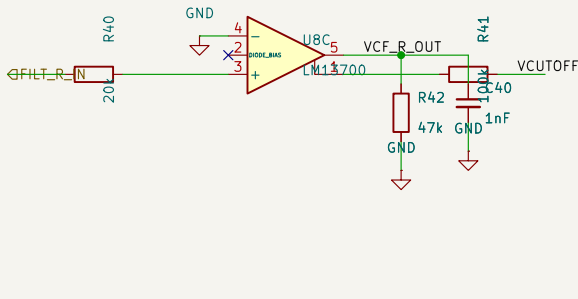
CV SUMMER – DAC CV_CUTOFF + CV_MOD summed by U5 -> VCUTOFF (drives both L and R VCFs); knob lives on the control board



VCF+VCA L – LM13700 OTA lowpass then OTA VCA (NOTE: OTA/labc topology – verify vs datasheet)



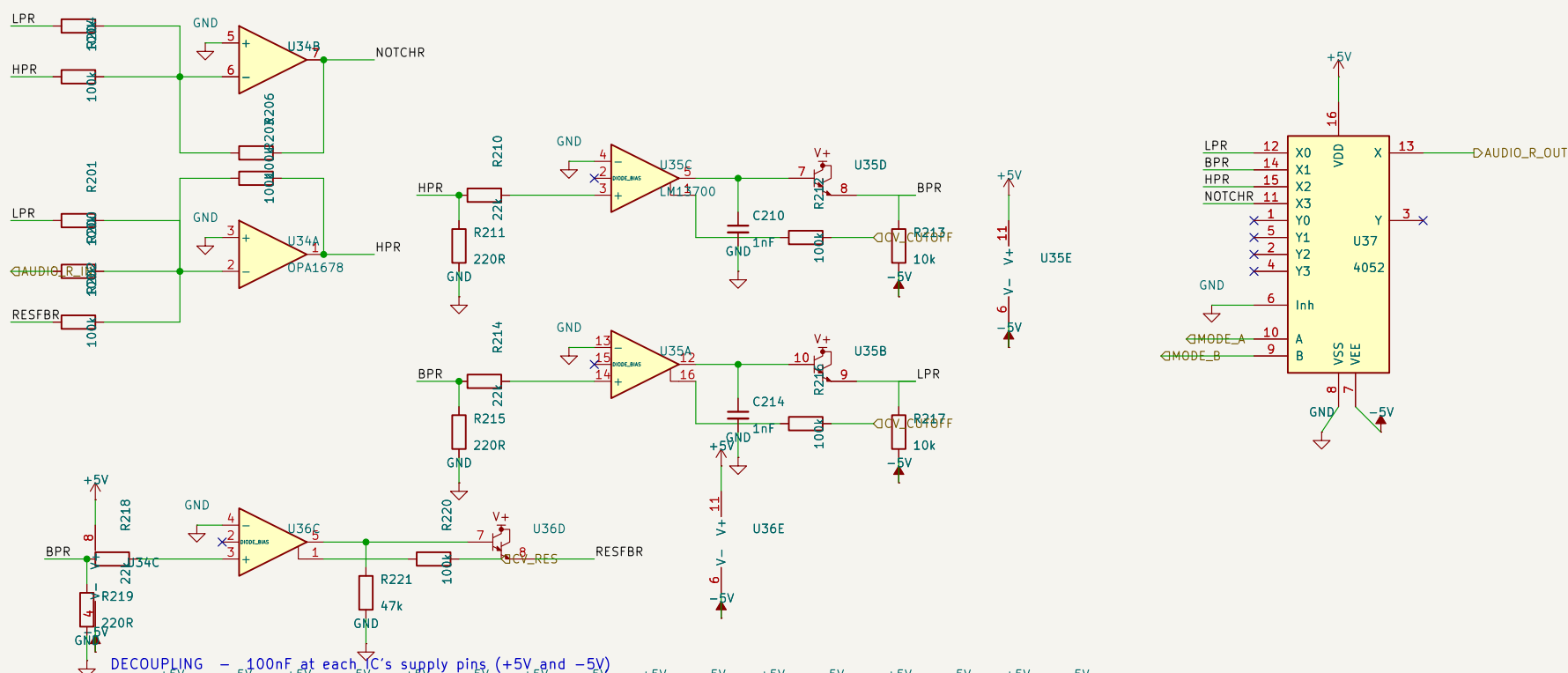
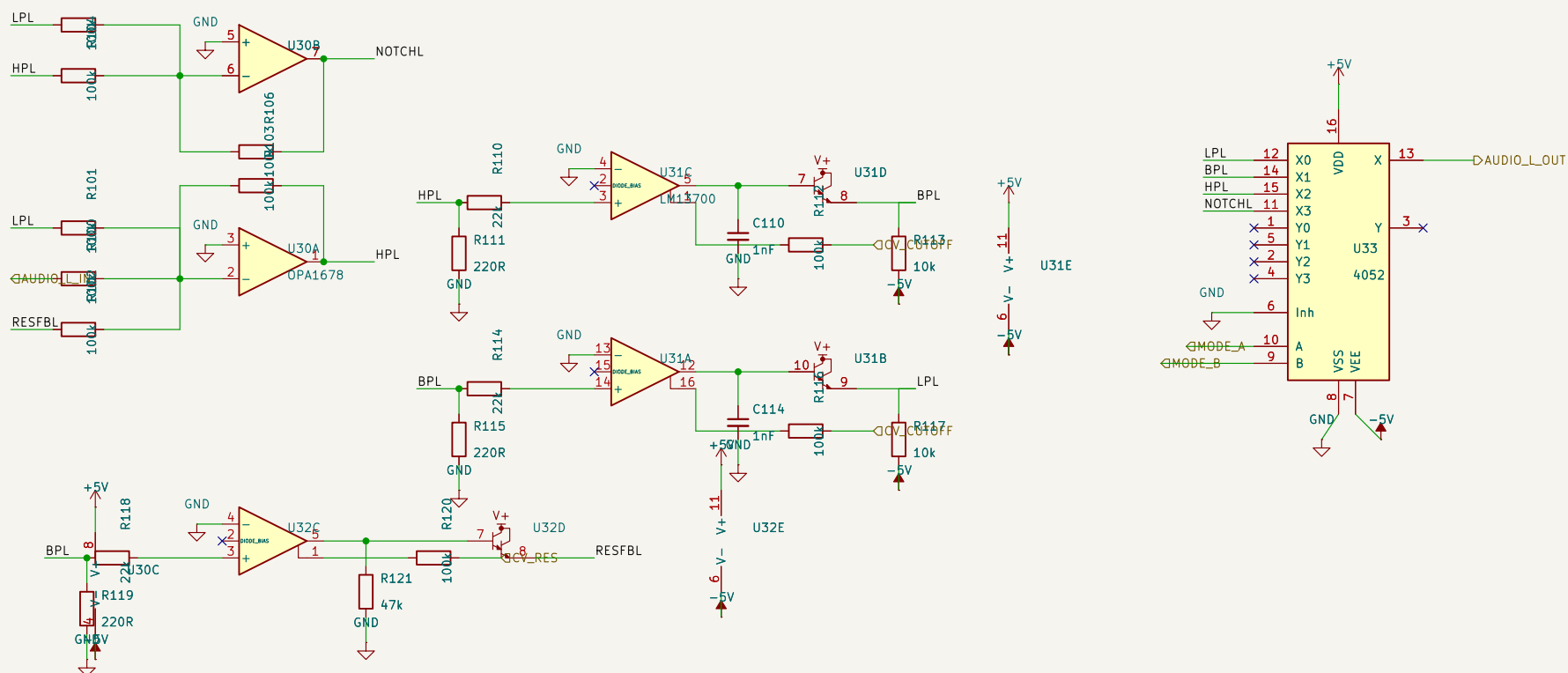
VCF+VCA R – LM13700 OTA lowpass then OTA VCA (NOTE: OTA/labc topology – verify vs datasheet)



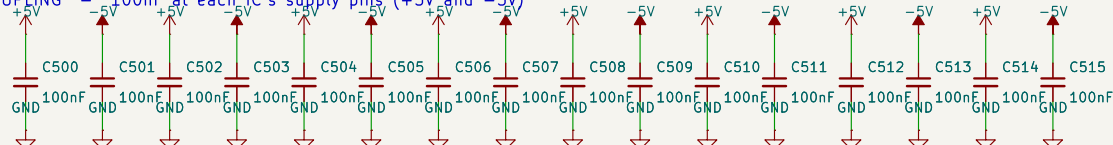
FILT_x_IN / FILT_x_OUT are the hierarchical pins shared with the root (front-end) sheet. Power rails (+5V/-5V/GND) are global.

Sheet: /ANALOG_FILTER_CV/ File: ANALOG_FILTER_CV.kicad_sch		
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MASTER FILTER (stereo) - VC state-variable filter 2-pole: LP/HP/BP/notch; CV cutoff + CV resonance; mode-select mux



DECOUPLING - 100nF at each IC's supply pins (+5V and -5V)



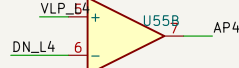
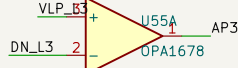
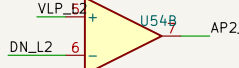
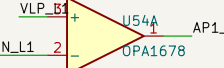
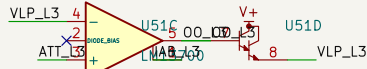
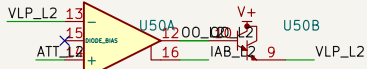
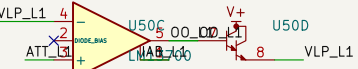
NOTE: OTA VC-SVF¹ + OTA resonance VCA² — prototype/verify levels, labc, buffer bias vs LM13700 datasheet before fab. ORDER_SEL reserved for the 2/4-pole cascade.⁴

Sheet: /MASTER_FILTER/ File: MASTER_FILTER.kicad_sch		
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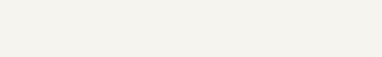
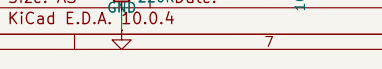
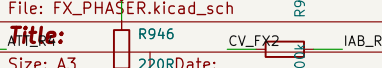
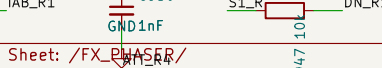
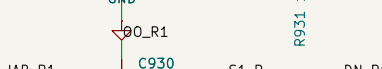
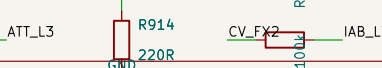
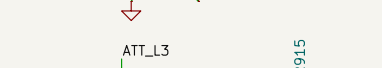
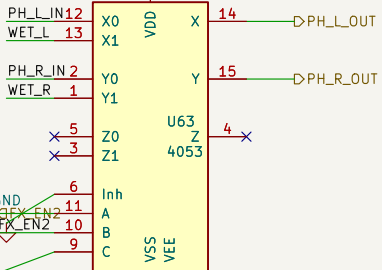
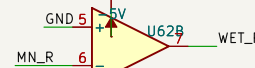
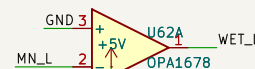
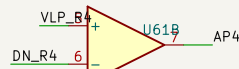
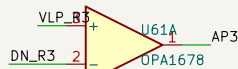
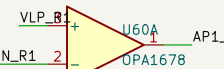
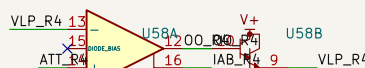
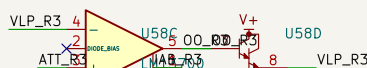
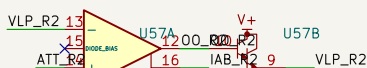
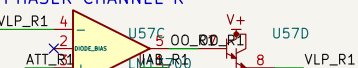
FX PHASER (true stereo) – 4 OTA all-pass stages/channel swept by CV_FX2 (STM32 LFO); feedback for notch depth; dry+wet mix; bypass via CD4053 on FX_EN2.

all-pass stage = OTA-C lowpass (Vlp) + diff amp (2*Vlp – Vin). NOTE: connectivity + topology verified; LEVELS/OTA-bias need real-LM13700-model sim before fab.

PHASER CHANNEL L



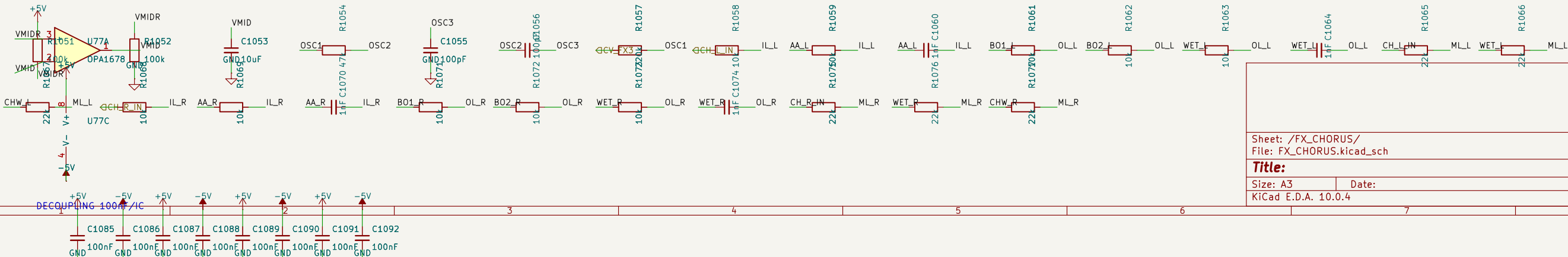
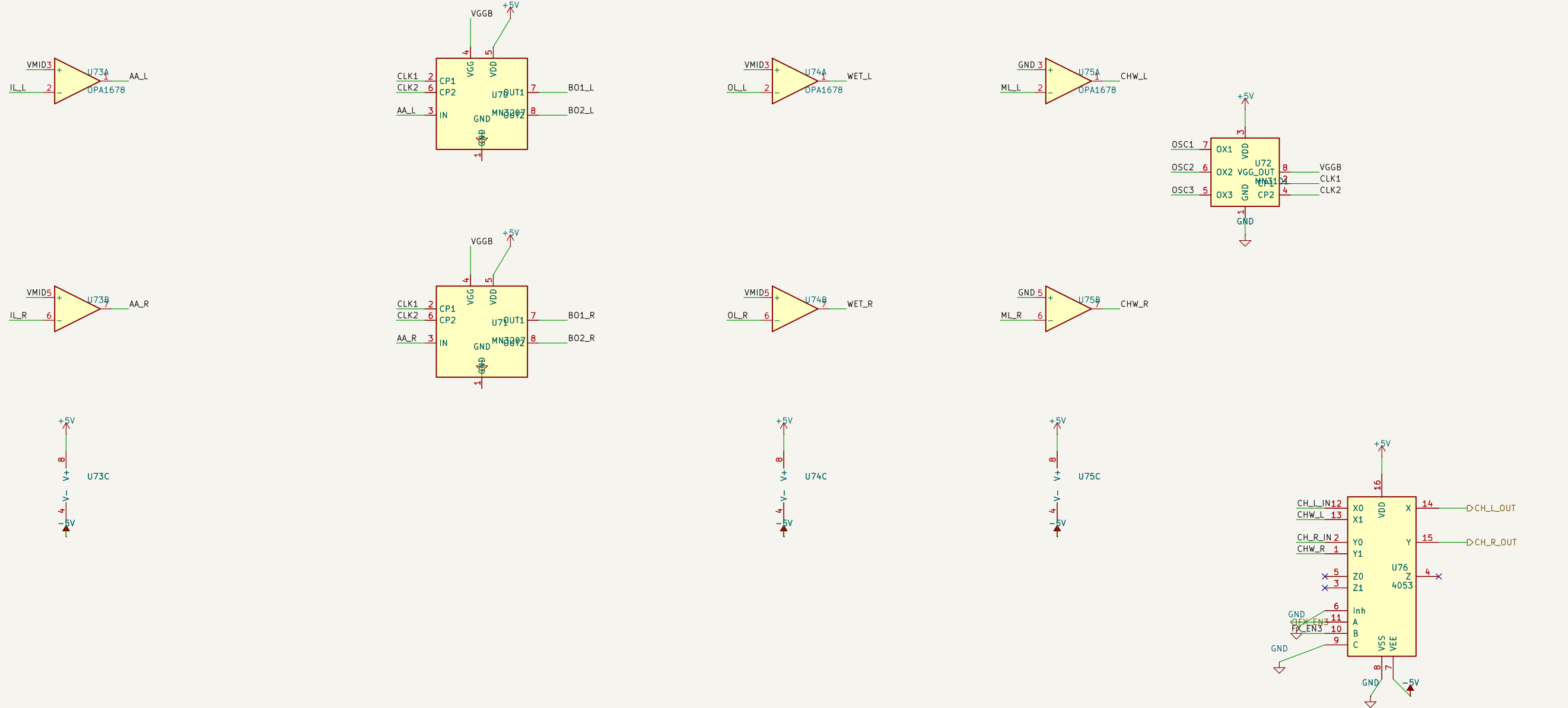
PHASER CHANNEL R



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FX CHORUS (true stereo) - BBD (MN3207) delay per channel, MN3101 2-phase clock swept by CV_FX3 (vibrato); wet/dry mix; bypass CD4053 on FX_EN3.

Anti-alias LPF in, reconstruction LPF out. NOTE: NO NE570 compander (not in KiCad libs) - add later for lower BBD noise. VDD=+5V (verify BBD headroom/bias + clock rate vs datasheet + real-model sim before fab).



Sheet: /FX_CHORUS/ File: FX_CHORUS.kicad_sch		
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